

One gene, three scales of the Common Fund

by Aaron Kanzer · [interactive version at https://scigantic.com/sample/cfde-ret-three-scales](https://scigantic.com/sample/cfde-ret-three-scales)

The NIH Common Fund funds programs that each look at human biology through a different lens. This notebook follows a single gene, **RET**, across three of them, all open and all mounted live on Scigantic:

1. **4D Nucleome (4DN)** shows how the chromosome RET sits on physically folds in 3D.
2. **GTEx** shows where RET is switched on across 54 human tissues.
3. **SPARC** zooms all the way in to the individual gut cells that express it.

Why RET? It encodes the receptor for GDNF, the signal that builds the gut's own nervous system (the same enteric neurons SPARC profiles), and it is also a major cancer drug target. So one gene ties chromatin folding, tissue expression, and single-cell identity together.

- **4DN** `s3://4dn-open-data-public` · **GTEx** `gs://adult-gtex` · **SPARC** `s3://sparc-prod-aod-discover-publish50-use1`
- All public, all Common Fund, **no account needed**.

```
# Two datasets are FUSE-mounted into this pod (4DN, SPARC); GTEx is read over public HTTPS.
# cooler (Hi-C) and leiden (clustering) are not in the base image, so install them quietly.
%pip install -q cooler leidenalg igraph 2>/dev/null

import os, warnings, urllib.request, gzip
import numpy as np, pandas as pd
import matplotlib.pyplot as plt
warnings.filterwarnings("ignore")
plt.rcParams.update({"figure.dpi": 70, "savefig.dpi": 70, "font.size": 9})

# RET, GRCh38: chr10:43,077,096-43,130,351
RET_CHROM, RET_POS = "chr10", 43_100_000

def resolve_root(probe, extra=()):
    """Find the mount path that holds `probe`, across bake-pod and live-pod layouts."""
    candids = ["/mnt/archive", os.path.expanduser("~/archive"), *extra]
    for c in candids:
        if os.path.exists(os.path.join(c, probe)):
            return c
    return candids[0]

FOURDN_ROOT = resolve_root("fourfront-webprod",
                           extra=["/mnt/archives/4dn-open-data-public",
                                   os.path.expanduser("~/archives/4dn-open-data-public")])
SPARC_ROOT = resolve_root("files/derivative/human_aggregated_matrix.mtx",
                          extra=["/mnt/archives/sparc-prod-aod-discover-publish50-use1/222",
                                  os.path.expanduser("~/archives/sparc-prod-aod-discover-publish50-
use1/222")])
GTEx_URL = ("https://storage.googleapis.com/adult-gtex/bulk-gex/v8/rna-seq/"
            "GTEx_Analysis_2017-06-05_v8_RNASeQCv1.1.9_gene_median_tpm.gct.gz")
print("4DN :", FOURDN_ROOT)
print("SPARC:", SPARC_ROOT)
```

Note: you may need to restart the kernel to use updated packages.

```
4DN : /mnt/archive
SPARC: /mnt/archives/sparc-prod-aod-discover-publish50-use1/222
```

Scale 1: the chromosome, folded in 3D (4D Nucleome)

A genome is two meters of DNA packed into a nucleus a few microns wide. **4DN** measures that packing with Hi-C, which counts how often any two stretches of DNA touch. Below is chromosome 10, the chromosome RET sits on, shown as **observed/expected** contacts. Dividing out the trivial "near things touch" signal exposes the **A/B compartment** checkerboard: the red squares are active DNA preferentially touching other active DNA, the blue the inactive counterpart. The dashed lines mark RET's position, inside one of these folded compartments that govern when it can be switched on.

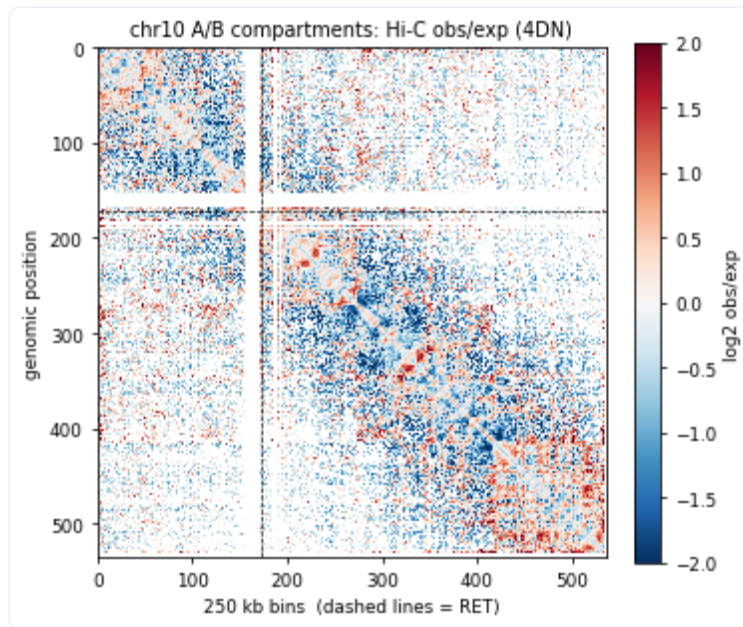
```
import cooler

mcool = os.path.join(FOURDN_ROOT, "fourfront-webprod", "wfoutput",
                    "00328169-edbc-4464-b48d-aa50cbe0f0cf", "4DNFIQ9FG7RR.mcool")
res_avail = sorted(int(u.rsplit("/", 1)[-1]) for u in cooler.fileops.list_coolers(mcool))
res = min([r for r in res_avail if r >= 250_000] or [max(res_avail)])
c = cooler.Cooler(f"{mcool}::resolutions/{res}")
chrom = next((x for x in (RET_CHROM, RET_CHROM[3:]) if x in c.chromnames), c.chromnames[0])
print(f"resolutions: {res_avail[:3]}...{res_avail[-2:]} | {chrom} @ {res//1000} kb")
try:    M = c.matrix(balance=True).fetch(chrom)
except Exception: M = c.matrix(balance=False).fetch(chrom)
M = np.nan_to_num(M)

# observed/expected: divide each diagonal by its mean -> reveals the A/B checkerboard
n = M.shape[0]
OE = np.full_like(M, np.nan, dtype=float)
for k in range(n):
    d = np.diagonal(M, k); mu = d[d > 0].mean() if (d > 0).any() else 0
    if mu > 0:
        i = np.arange(n - k); OE[i, i + k] = M[i, i + k] / mu; OE[i + k, i] = OE[i, i + k]
log2oe = np.log2(OE, where=(OE > 0), out=np.full_like(OE, np.nan))

fig, ax = plt.subplots(figsize=(5.4, 4.9))
im = ax.imshow(log2oe, cmap="RdBu_r", vmin=-2, vmax=2, interpolation="nearest")
rb = RET_POS // res
ax.axhline(rb, color="k", lw=0.7, ls="--"); ax.axvline(rb, color="k", lw=0.7, ls="--")
ax.set_title(f"{chrom} A/B compartments: Hi-C obs/exp (4DN)", fontsize=10)
ax.set_xlabel(f"{res//1000} kb bins (dashed lines = RET)"); ax.set_ylabel("genomic position")
plt.colorbar(im, ax=ax, fraction=0.046, label="log2 obs/exp")
plt.tight_layout(); plt.show()
```

```
resolutions: [1000, 2000, 5000]...[5000000, 10000000] | chr10 @ 250 kb
```



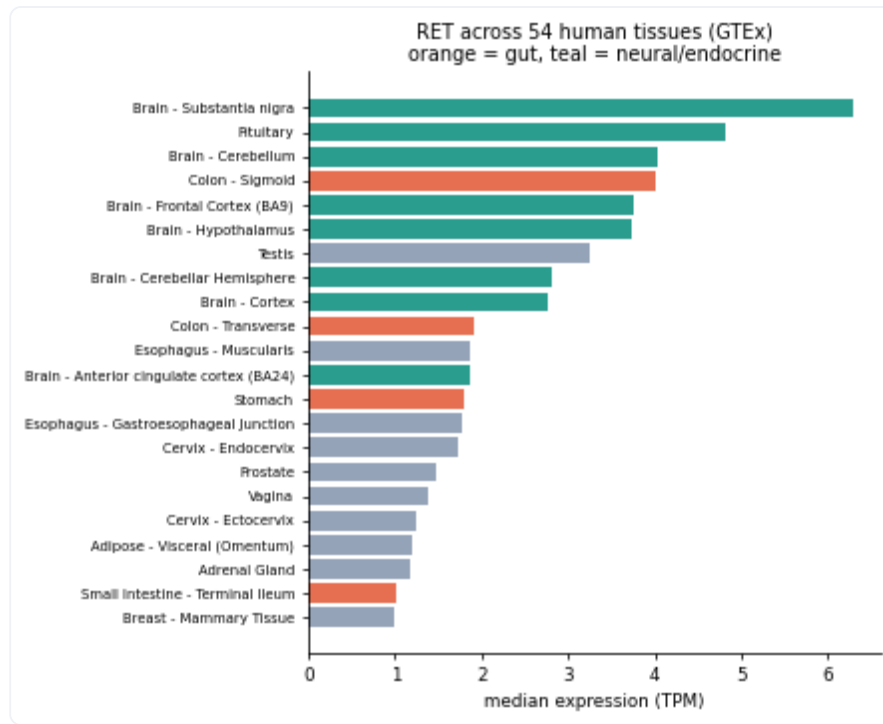
DNA does not fold randomly. That compartment is the physical neighborhood deciding when RET can be switched on. The next scale shows which tissues actually do.

Scale 2: one genome, 54 tissues (GTEx)

Every cell carries the same RET gene, but only some tissues read it. **GTEx** measured median expression for every gene across 54 human tissues from hundreds of donors. Streaming just RET's row out of the public GTEx matrix shows where it is actually expressed: the nervous system and endocrine glands, and notably the **colon**, the organ SPARC profiles next.

```
raw = urllib.request.urlopen(GTEX_URL, timeout=180).read()
lines = gzip.decompress(raw).decode().split("\n")
tissues = lines[2].split("\t")[2:]
ret_vals = None
for ln in lines[3:]:
    p = ln.split("\t")
    if len(p) > 1 and p[1] == "RET":
        ret_vals = [float(x) for x in p[2:]]; break

s = pd.Series(ret_vals, index=tissues).sort_values(ascending=False).head(22)[::-1]
def color(t):
    if "Colon" in t or "Intestine" in t or "Stomach" in t: return "#e76f51"    # gut
    if "Brain" in t or "Nerve" in t or "Pituitary" in t: return "#2a9d8f"    # neural/endocrine
    return "#94a3b8"
fig, ax = plt.subplots(figsize=(6.4, 5.2))
ax.barh(range(len(s)), s.values, color=[color(t) for t in s.index])
ax.set_yticks(range(len(s))); ax.set_yticklabels(s.index, fontsize=7)
ax.set_xlabel("median expression (TPM)")
ax.set_title("RET across 54 human tissues (GTEx)\norange = gut, teal = neural/endocrine", fontsize=10)
for sp in ("top", "right"): ax.spines[sp].set_visible(False)
plt.tight_layout(); plt.show()
```



Scale 3: one organ, single cells (SPARC)

Tissue-level expression still averages over many cell types. **SPARC** profiled the colon's own nervous system one nucleus at a time. We rebuild that single-cell map (same human colon matrix as the standalone SPARC notebook), cluster it, and look for RET. RET is a receptor, so its transcript is sparse in single nuclei, but it concentrates in the **neuron compartment** (marked by ELAVL4), exactly where GDNF/RET signaling acts to build the gut's nervous system.

```

import scanpy as sc, anndata as ad
from scipy.io import mmread
sc.settings.verbosity = 0
sc.settings.set_figure_params(dpi=54, dpi_save=54, frameon=False, figsize=(3.9, 3.5))

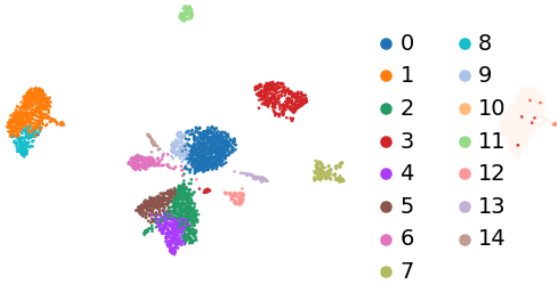
base = os.path.join(SPARC_ROOT, "files", "derivative", "human_aggregated")
M = mmread(base + "_matrix.mtx").tocsr()
genes = pd.read_csv(base + "_genes.tsv", sep="\t", header=None)
bc = pd.read_csv(base + "_barcodes.tsv", sep="\t", header=None)
adata = ad.AnnData(X=M.T.tocsr())
adata.obs_names = bc[0].astype(str).values
adata.var_names = genes[1].astype(str).values
adata.var_names_make_unique()

sc.pp.filter_genes(adata, min_cells=3)
if adata.n_obs > 3500:
    sc.pp.subsample(adata, n_obs=3500, random_state=0)
sc.pp.normalize_total(adata, target_sum=1e4); sc.pp.log1p(adata); adata.raw = adata
hv = adata[:, sc.pp.highly_variable_genes(adata, n_top_genes=2000, inplace=False)
["highly_variable"].values].copy()
sc.pp.scale(hv, max_value=10); sc.tl.pca(hv, n_comps=30)
sc.pp.neighbors(hv, n_neighbors=15, n_pcs=30); sc.tl.umap(hv)
try:
    sc.tl.leiden(hv, resolution=1.0, key_added="cluster")
except Exception:
    from sklearn.cluster import KMeans
    hv.obs["cluster"] = pd.Categorical(KMeans(12, n_init=10,
random_state=0).fit_predict(hv.obsm["X_pca"]).astype(str))
adata.obsm["X_umap"] = hv.obsm["X_umap"]
adata.obs["cluster"] = hv.obs["cluster"].values

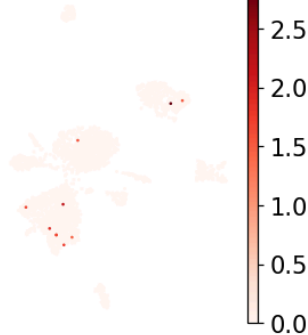
color = ["cluster"] + [g for g in ["RET", "ELAVL4"] if g in adata.raw.var_names]
# Reds (white at zero) so the sparse RET / ELAVL4 positive nuclei pop instead of
# vanishing into a dark viridis background.
sc.pl.umap(adata, color=color, ncols=3, size=12, cmap="Reds",
          title=["Colon cell clusters (SPARC)", "RET", "ELAVL4 (neurons)"])

```

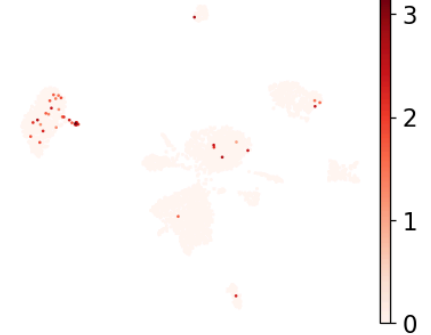
Colon cell clusters (SPARC)



RET



ELAVL4 (neurons)



What you just saw

One gene, **RET**, seen at three scales of biology, each from a different NIH Common Fund program, each mounted live on Scigantic with no download:

- **4DN** gave us the folded chromosome it lives on.
- **GTEx** gave us the tissues that express it (gut and nervous system).
- **SPARC** gave us the individual enteric neurons that switch it on.

The same molecule, from chromatin contacts to a single nucleus. This is what the **Common Fund Data Ecosystem (CFDE)** is for: programs that each measure biology differently, made interoperable so one question can cross all of them.

- **4DN** Hi-C contact matrix (`4dn-open-data-public`) · **GTEx** v8 median TPM (`adult-gtex` , GCS) · **SPARC** dataset 222 (`sparc-prod-aod-discover-publish50-use1`)
- Licenses: 4DN CC-BY-4.0 · GTEx open-access · SPARC CC-BY-4.0